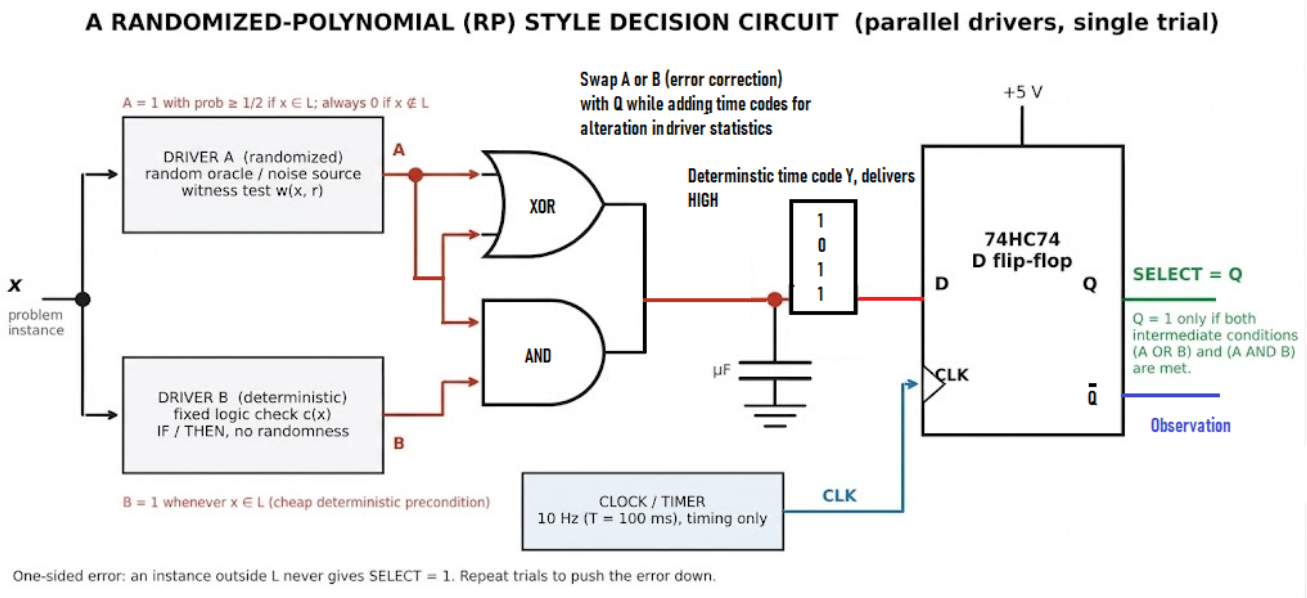
How the RP-Style Decision Circuit Works

A walkthrough of the parallel-driver, single-trial circuit (A, B, XOR/AND, time code Y, 74HC74 flip-flop)



The circuit as supplied

1. The big idea

The circuit is a hardware picture of a **Randomized Polynomial-time (RP)** decision procedure. RP is the class of yes/no problems for which a probabilistic algorithm has **one-sided error**: if the answer is **NO** (x is not in L) the algorithm *never* says yes, and if the answer is **YES** (x is in L) it says yes with probability at least 1/2. The circuit runs one such trial. It takes the problem instance **x**, runs it through two drivers in parallel, combines their outputs with logic gates, and stores the verdict in a flip-flop on a clock edge. That stored bit, **SELECT = Q**, is the answer.

2. The blocks and what each one does

Block	Role	Behaviour
Driver A (randomized)	Probabilistic witness test	Uses a random oracle / noise source to try a random witness r and test w(x, r). If x is in L, A = 1 with probability ≥ 1/2. If x is not in L, A = 0 always (no witness can exist).
Driver B (deterministic)	Cheap precondition	A fixed IF/THEN check c(x) with no randomness. B = 1 whenever x is in L. It acts as a sanity filter that costs almost nothing.
XOR gate	“Swap / error-correction” branch	Outputs 1 when A and B disagree. In the figure's notes it is the stage that swaps A or B and adds time codes to alter the driver statistics.
AND gate	Agreement check	Outputs 1 only when A = 1 and B = 1, i.e. the random witness test succeeded *and* the deterministic check agrees.
Capacitor (μF) node	Smoothing / hold	The gate outputs meet at a node with a capacitor to ground. It low-pass filters glitches so the signal is stable when the clock samples it.
Time code Y = 1011	Deterministic time code	A fixed bit pattern that delivers a HIGH to the D input, giving the stage a known, repeatable timing signature.
Clock / timer	Sampling instant	10 Hz (T = 100 ms), “timing only”. It carries no data, it just tells the flip-flop when to look at D.

Block	Role	Behaviour
74HC74 D flip-flop	Memory of the verdict	On the rising CLK edge it copies D to Q and holds it. Q is SELECT (the decision). Q̄ (Q-bar) is the inverted output, labelled Observation.

3. Step-by-step signal flow

1. **Input.** The instance x is fanned out to both drivers at once (the black dot at the left).
2. **Drivers fire in parallel.** A flips its random coin(s) and tests a witness. B runs its fixed check. Both produce one bit, labelled A and B in red.
3. **Gates combine them.** A and B go to the XOR and AND gates. The caption summarises the requirement as two intermediate conditions: (A OR B) and (A AND B).
4. **Node + capacitor.** The gate outputs join at one node. The capacitor smooths it and the time code Y presents a defined HIGH to the flip-flop's D pin.
5. **Clock edge.** Every 100 ms the rising edge on CLK makes the 74HC74 latch whatever is on D.
6. **Result.** Q = SELECT = 1 means “accept: x is in L”. Q = 0 means “no witness found”. Q-bar shows the opposite polarity for observation.

4. Why the error is one-sided

The caption says Q = 1 only if both conditions hold, which collapses to **A AND B** (since A AND B already implies A OR B). So the circuit accepts only when the randomized driver finds a witness:

Case	A	B	SELECT (Q)	Meaning
x not in L	always 0	any	0	Never a false accept. No error possible.
x in L, witness found	1	1	1	Correct accept (probability ≥ 1/2).
x in L, witness missed	0	1	0	False reject (probability ≤ 1/2). The only error type.

This matches the footer of the diagram: *an instance outside L never gives SELECT = 1*. A wrong answer can only be a missed YES, never a fake YES.

5. Repeating trials to push the error down

One trial can wrongly reject a YES instance with probability up to 1/2. Because trials are independent and a single success is conclusive, you run the circuit k times and accept if *any* trial gives SELECT = 1. The chance of every trial failing is at most $(1/2)^k$. At the 10 Hz clock, k trials take k × 100 ms.

Trials k	Worst-case error	Time at 10 Hz
1	0.5	0.1 s
2	0.25	0.2 s
5	0.03125	0.5 s
10	0.000976562	1.0 s
20	9.54e-07	2.0 s

6. Things to be aware of in the drawing

The figure is a conceptual illustration, not a build-ready schematic. If you ever wire it up, check these points:

- **Mutually exclusive gates.** XOR and AND can never both output 1 for the same inputs. So a literal “XOR AND AND” is always 0, while the two outputs simply tied together behave like OR (A OR B). The caption's (A OR B) and (A AND B) reading is what keeps the logic meaningful, and it

reduces to A AND B.

- **Tied gate outputs.** Directly joining two CMOS outputs causes bus contention when they disagree. A real design would use an OR gate (or a proper combining stage) instead.
- **A OR B alone would break the guarantee.** If B can be 1 for an x outside L, accepting on A OR B could give a false accept. Only the AND path preserves one-sided error.
- **74HC74 housekeeping.** Real use needs the PRE and CLR pins tied high, VCC and GND connected, and decoupling near the chip. These are not shown.
- **Q-bar is not extra information.** It is simply NOT Q, useful for an indicator LED or active-low logic.

Summary: two drivers test x in parallel, the gates demand agreement, a clocked D flip-flop latches the verdict, and repetition drives the false-reject rate toward zero while false accepts stay impossible.